Cairo University
Faculty of Engineering
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RSSI — Remote Sensing

Course Project

24 Mar 2026 • Land Cover

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A demanding road for the spirit

PROJECT OVERVIEW

The framework for the Land Cover Classification project is established within this document. Both classical machine learning and deep learning techniques are to be applied to perform multi-class semantic segmentation on high-resolution Sentinel-2 satellite imagery. Results are expected to be critically analyzed, multi-band spectral data must be handled, and the practical implications of these findings in remote sensing applications should be thoroughly understood.

OBJECTIVE

Classical and deep learning methods for segmenting satellite images into specific land cover classes are to be implemented, benchmarked, and compared, with the overarching goal of mapping complex terrains accurately.

DATASET DESCRIPTION

A curated dataset of Sentinel-2 Level-2A imagery sourced from diverse Egyptian terrains (e.g., dense urban centers, desert oases, and agricultural boundaries) will be provided for this task.

- **Input Data:** 12-band continuous spectral data (GeoTIFF format) and 3-band RGB visual references will be supplied.
- **Ground Truth:** Masked images representing 5 distinct classes are utilized:
  - `0`: Unknown (Black)
  - `1`: Greenery (Dark Green)
  - `2`: Sand (Yellow)
  - `3`: Water (Dark Blue)
  - `4`: Cement/Infrastructure (Grey)

Important Note on the 'Unknown' Class

The 'Unknown' class (`0`) is provided purely as a dataset safeguard for unclassified, boundary, or corrupted pixels. Models are **not** expected to be trained to predict the 'Unknown' class, and it will be excluded entirely from the final evaluation metrics.

DATASET ACQUISITION AND MASK GENERATION

Massive amounts of annotated data are required to successfully train deep learning semantic segmentation architectures. Whether external datasets are downloaded or custom data is generated, it must be ensured that all ground-truth labels are mapped precisely to the project's 5-class structure: `0`: Unknown (safeguard/ignore), `1`: Greenery, `2`: Sand, `3`: Water, and `4`: Cement/Infrastructure.

Script-Based Data Extraction (Recommended)

For a robust and scalable approach, scripts can be written to sample from planetary-scale land cover maps hosted on platforms such as Google Earth Engine.

- **Dynamic World (Google & WRI):** A near real-time global dataset featuring massive hand-annotated training data (`https://dynamicworld.app/`). *Class Mapping:* Water → Water, Trees/Grass/Crops → Greenery, Built Area → Cement, Bare Ground → Sand.
- **Esri 10m Annual Land Cover:** A comprehensive dataset utilizing 9 primary classes (`https://gee-community-catalog.org/projects/S2TSLULC/`). *Class Mapping:* Water → Water, Trees/Crops/Rangeland → Greenery, Built Area → Cement, Bare Ground → Sand.

Benchmark Machine Learning Datasets

Alternatively, pre-cut, ready-to-train pairs of Sentinel-2 images and ground-truth masks can be downloaded from the following open-source benchmark datasets:

- **LandCoverNet (Radiant Earth):** High-quality 256×256 multi-spectral image chips paired with human-verified semantic masks. (`https://source.coop/radiantearth/landcovernet`)
- **SEN12MS Dataset:** Over 180,000 pre-processed patch triplets of Sentinel-2 multi-spectral images paired with aligned annotations. (`https://mediatum.ub.tum.de/1474000`)
- **Sen-2 LULC Dataset (2023):** Over 213,000 pre-processed Sentinel-2 RGB images and corresponding masks, pre-divided into train, test, and validation sets. (`https://data.mendeley.com/datasets/f4ky6ks248/3`)

Custom Mask Generation via Python Tooling

For precise, localized ground-truth generation and test-set validation, a custom Python ground-truth editor is provided. Raw 12-band arrays can be loaded into this tool so polygon boundaries can be manually constructed or refined over specific Egyptian terrains. Features such as dual-synced crosshairs for RGB preview tracing, precision vertex undo capabilities, and rapid reclassification shortcuts are included to guarantee that custom masks perfectly align with the target multi-class structure.

Retrieving Spectral Inputs via Google Earth Engine

To acquire the required continuous 12-band spectral imagery for testing and extending the dataset, the following JavaScript snippet can be executed in the Google Earth Engine Code Editor. This code filters, cloud-masks, and exports the required Sentinel-2 spectral imagery directly to Google Drive.

```
// Define the Google Drive Folder Name
var DRIVE_FOLDER = 'S2_ML_Dataset';

// Define selected target locations
var locations = [
  {name: 'CairoUniv', lon: 31.20909, lat: 30.02633},
  {name: 'IconicTower',  lon: 31.7018369, lat: 30.0109181}, // New Admin Capital CBD
  {name: 'SiwaOasis',    lon: 25.5195, lat: 29.2032}, // Central Siwa / Shali Fortress
  {name: 'KarnakLuxor',  lon: 32.6583, lat: 25.7180}, // Karnak Temple Complex
  {name: 'PhilaeAswan',  lon: 32.8844, lat: 24.0255}, // Philae Temple on the Nile
  {name: 'HawaraFayoum', lon: 30.8986, lat: 29.2747}  // Labyrinth / Pyramid of Hawara
];

// Most recent summer to avoid clouds
var startDate = '2025-05-01';
var endDate = '2025-08-28';
```

```
// Cloud Masking Function
function maskClouds(img) {
  var scl = img.select('SCL');
  var mask = scl.neq(3).and(scl.neq(8)).and(scl.neq(9)).and(scl.neq(10));
  return img.updateMask(mask);
}

// ========================================
// MAIN PROCESSING LOOP
// ========================================

locations.forEach(function(loc) {

  var region = ee.Geometry.Point([loc.lon, loc.lat]).buffer(1500).bounds();

  var s2_sr = ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED")
    .filterBounds(region)
    .filterDate(startDate, endDate)
    .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20));

  var clean_col = s2_sr.map(maskClouds);

  var target_bands = ['B1', 'B2', 'B3', 'B4', 'B5', 'B6', 'B7', 'B8', 'B8A', 'B9', 'B11'
      , 'B12'];
  var clean_l2a = clean_col.select(target_bands).median().clip(region);

  // ========================================
  // CREATE GOOGLE DRIVE EXPORT TASKS
  // ========================================

  // Task: Export Full Spectral Data (12 Bands)
  Export.image.toDrive({
    image: clean_l2a,
    description: loc.name + '_Spectral_300px',
    folder: DRIVE_FOLDER,
    scale: 10,
    region: region,
    maxPixels: 1e8,
    fileFormat: 'GeoTIFF'
  });

  print('Tasks generated for:', loc.name);
});

print('Check the "Tasks" tab to submit your streamlined exports!');
```

PROJECT REQUIREMENTS

- **Preprocessing:** Suitable data normalization and preprocessing must be applied to the 12-band 16-bit arrays (handling ranges from 0 to 10,000) to prepare them for model training.
- **Classical Technique:** At least one classical pixel-based classification or clustering technique (e.g., Random Forest, Support Vector Machines, or K-Means with feature engineering) must be implemented and documented.
- **Deep Learning Technique:** At least one deep learning architecture suited for semantic segmentation (e.g., U-Net, DeepLab, FCN, or Transformer-based models) must be implemented and documented.

PROJECT TIMELINE

- **Project Launch:** 24 Mar 2026
- **Test Set Release:** Images similar to the samples will be provided for evaluation prior to the deadline.
- **Final Project Submission:** 26 Apr 2026
- **Grades Announcement:** 27 Apr 2026

GRADING BREAKDOWN (15 MARKS TOTAL)

Because this is a multi-class segmentation task, submissions will be evaluated comprehensively. The overall grade will be distributed as follows:

Component	Marks	Evaluation Focus
Classical Approach	**/4**	Implementation quality, preprocessing enhancements, and baseline performance of the chosen classical machine learning model.
Deep Learning	**/4**	Architecture selection, training methodology, hyperparameter optimization, and output quality of the deep learning model.
Discussion	**/2**	Thoughtful analysis comparing both approaches, discussing computational trade-offs, edge cases, and architectural reasoning.
mIoU Performance	**/5**	Evaluated competitively based on the highest Mean Intersection over Union (mIoU) achieved across the 4 active target classes (Greenery, Sand, Water, Cement) on the withheld test set.

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c}$$

Where:

- TP_c (True Positives): Pixels correctly predicted as class c.
- FP_c (False Positives): Pixels incorrectly predicted as class c.
- FN_c (False Negatives): Pixels of class c incorrectly predicted as another class.

The mIoU is the average of the IoU scores across all N active classes. For this project, $N = 4$ (Greenery, Sand, Water, Cement). You must explicitly exclude the 'Unknown' class (`0`) from this average, also report the individual IoU scores for each class:

$$mIoU = \frac{1}{4}\sum_{c=1}^{4} IoU_c$$